openbuoyancy

Will it float, and will it stay upright? Archimedes in four sessions.

PAN IIT OSS explainer • higher secondary to diploma / undergraduate (levels 4-5)

Why this exists

Every ship, raft and balloon floats because of one idea of Archimedes, and every ship stays upright or rolls over because of where its weight sits relative to its metacentre. This explainer shows the mathematics with a box-shaped barge, so you can build it yourself.

The end goal

Decide whether a ship floats and stays upright using Archimedes' principle. The end goal is to decide whether a vessel floats and stays upright using Archimedes' principle. The same reasoning is behind life jackets, submarines, hot-air balloons and the loading rules for real ships, although those use far more detailed methods than this teaching model.

How the four sessions build up

1. **Session 1: Things float when the water they push aside weighs as much as they do** (Level 1, conceptual interactive model, 60 min).
2. **Session 2: The buoyant force, barge loads and balloon lift** (Level 2, mathematical engineering model, 75 min).
3. **Session 3: Does it stay upright? Metacentric height and the righting arm** (Level 3, simulation and validation, 90 min).
4. **Session 4: The free-surface effect: why a liquid in a tank lowers GM** (Level 3, simulation and validation, 90 min).

1 Session 1: Things float when the water they push aside weighs as much as they do

A. Real-world problem

A steel ship weighs thousands of tonnes and floats. A pebble weighs a few grams and sinks. A wooden block floats low in a pond and a little higher in the sea. Why? Can you predict, before you try, whether something floats and how deep it will sit? (This is general science education, not a guide for boat safety.)

B. Concept and theory

When an object is lowered into water it pushes some water out of its way. The water pushes back with a force equal to the weight of the water that was pushed aside (Archimedes' principle). An object floats when it can push aside its own weight in water before it is fully under. Because water displaced = fluid density times submerged volume, an object floats when its average density (total mass divided by total outside volume, including any air inside) is below the fluid's density, and then a fraction rho_object / rho_fluid of its volume is below the surface. For a flat-bottomed box

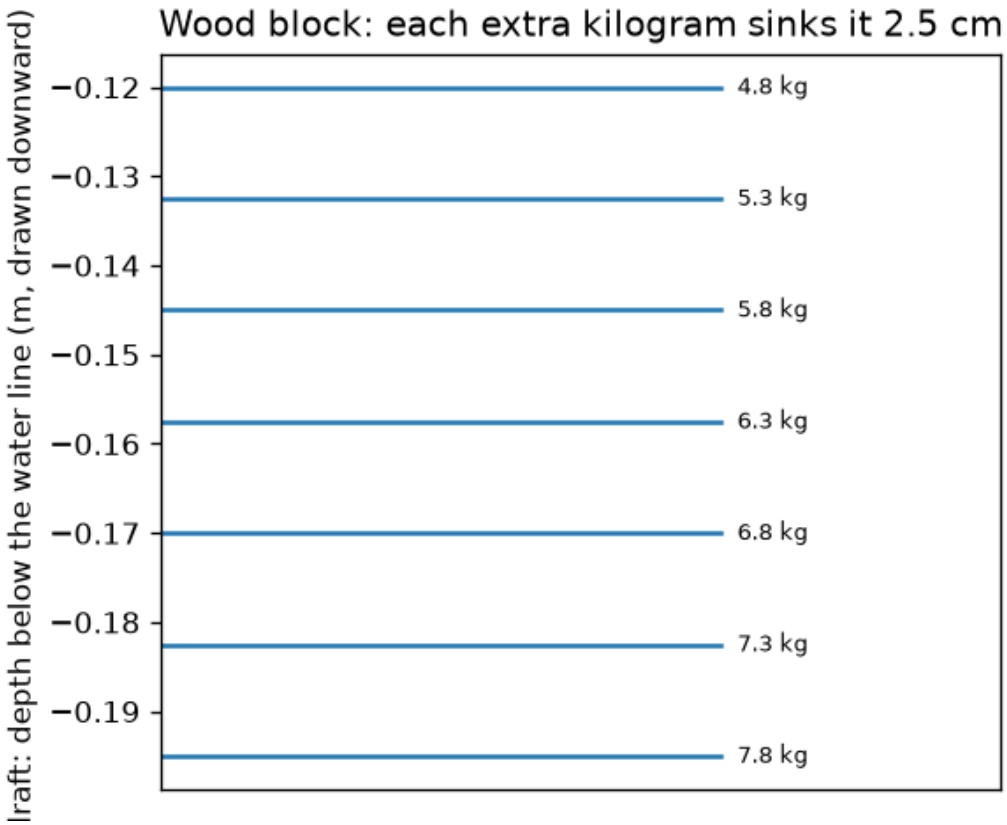
with bottom area A, the depth under water (the draft) is mass / (rho_fluid * A). A hollow steel boat floats because its average density includes the air inside.

- rho_obj: average density of the object (mass / outside volume) (kg/m^3)
- rho_fluid: density of the fluid (fresh water 1000, sea water about 1025) (kg/m^3)
- f: fraction of the object's volume under water (no unit)
- m: mass of the object (kg)
- A: area of the flat bottom (m^2)
- draft: depth of the bottom below the water line (m)

$$\rho_{obj} = \frac{m}{V}$$

$$f = \min\left(1,\ \frac{\rho_{obj}}{\rho_{fluid}}\right)$$

$$\text{draft} = \frac{m}{\rho_{fluid}\, A}$$



Draft marks of a wooden block: each extra kilogram adds 2.5 cm of draft.

E. Worked example

A wooden cube of side 0.2 m has mass 4.8 kg. Does it float in fresh water? How deep does it sit? What happens in sea water (1025 kg/m^3)?

1. Volume = 0.2 * 0.2 * 0.2 = 0.008 m^3, so average density = 4.8 / 0.008 = 600 kg/m^3.
2. 600 is below 1000, so it floats. Fraction under water = 600 / 1000 = 0.60.

3. Bottom area = 0.2 * 0.2 = 0.04 m^2, so draft = 4.8 / (1000 * 0.04) = 0.120 m (60 percent of 0.2 m, as it must be).
4. In sea water: draft = 4.8 / (1025 * 0.04) = 0.117 m. It sits about 3 mm higher.

Answer. It floats with 60 percent under water, draft 0.120 m in fresh water and 0.117 m in sea water.

F. Questions

1. **Conceptual.** Steel has a density of about 7850 kg/m^3, yet a steel ship floats. Explain, and say what would happen to the same steel hammered into a solid lump.
2. **Numerical.** A block of mass 2.0 kg has a volume of 0.0030 m^3. Find its average density, whether it floats in fresh water, the fraction under water and the mass of water it pushes aside.
3. **Graph and simulation interpretation.** In the graph of submerged fraction against density, the line stops rising at density 1000 (fresh water). What does an object of density 1010 kg/m^3 do in fresh water and in sea water?
4. **Design and optimisation challenge.** Design: a flat-bottomed boat has an empty mass of 150 kg and a bottom of 2.0 m by 0.8 m. In calm fresh water it must keep a draft of no more than 0.25 m. What is the largest cargo? State your assumptions.

2 Session 2: The buoyant force, barge loads and balloon lift

A. Real-world problem

A river barge is 20 m long and 6 m wide. How much cargo can it carry before its deck goes under water? Does the answer change when it sails out to sea? And how can a balloon rise in air with the same law? (Educational model: real vessels need larger margins, rules and professional design.)

B. Concept and theory

The buoyant force on a body equals the weight of the fluid it displaces: F_b = rho_fluid g V_displaced. A body floats in equilibrium when this equals its weight m g. For a rectangular barge of length L, beam B and hull depth H, the displaced volume is L B d, so the draft is d = m / (rho L B) and the freeboard (deck height above the water) is H - d. The deck reaches the water at d = H, so the greatest total mass is rho L B H, and the load capacity is that minus the empty (lightship) mass; designers keep a minimum freeboard f_min, which lowers it to rho L B (H - f_min) - m_lightship. Air is also a fluid: a balloon filled with a gas of density rho_gas in air of density rho_air experiences a net lift of (rho_air - rho_gas) g V, so the lift per cubic metre is the density difference. The density of an ideal gas is rho = P M / (R T).

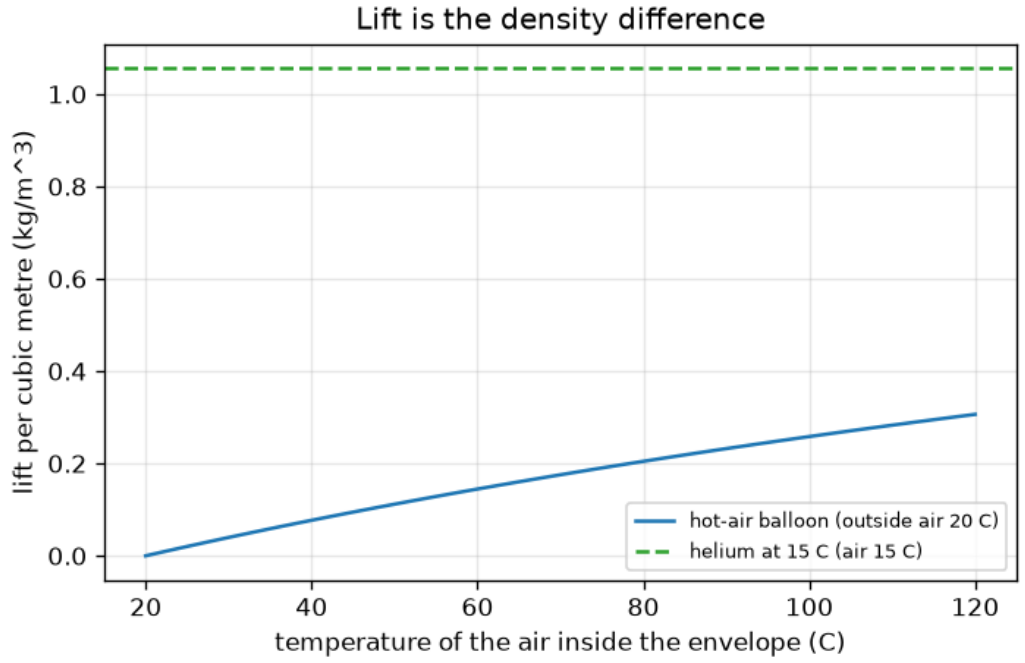
- F_b: buoyant force (N)
- rho: density of the fluid (kg/m^3)
- g: gravity, 9.81 (m/s^2)
- V: displaced volume (m^3)
- L, B, H: barge length, beam (width), hull depth (m)
- d: draft (m)
- m: total mass of barge and cargo (kg)
- P, M, R, T: gas pressure (Pa), molar mass (kg/mol), gas constant 8.314 J/(mol K), temperature (K) (as stated)

$$F_b = \rho_{fluid}\, g\, V_{displaced}$$

$$d = \frac{m}{\rho\, L\, B} \qquad \text{freeboard} = H - d$$

$$m_{cargo,max} = \rho\, L\, B\, (H - f_{min}) - m_{lightship}$$

$$\text{lift per m}^3 = \rho_{air} - \rho_{gas} \ \ (\text{kg/m}^3), \qquad \rho = \frac{P\, M}{R\, T}$$



Lift per cubic metre of a hot-air balloon against the inside temperature, with helium for comparison.

E. Worked example

A barge 20 m long, 6 m wide with hull depth 2.5 m and empty mass 60 t carries 100 t in fresh water. Find the draft and freeboard, the greatest cargo with the deck just awash, and the greatest cargo if 0.3 m of freeboard must remain.

1. Total mass = 60 000 + 100 000 = 160 000 kg. Draft = 160 000 / (1000 * 20 * 6) = 1.333 m.
2. Freeboard = 2.5 - 1.333 = 1.167 m.
3. Deck awash: greatest total mass = 1000 * 20 * 6 * 2.5 = 300 000 kg, so cargo = 300 000 - 60 000 = 240 000 kg = 240 t.
4. With 0.3 m freeboard: 1000 * 20 * 6 * (2.5 - 0.3) = 264 000 kg total, so cargo = 204 t.
5. Check with the buoyant force: at 1.333 m draft the displaced volume is 160 m^3 and F_b = 1000 * 9.81 * 160 = 1.57 MN, equal to the weight 160 000 * 9.81 N.

Answer. Draft 1.333 m, freeboard 1.167 m, capacity 240 t with the deck awash and 204 t with 0.3 m freeboard.

F. Questions

1. **Conceptual.** The barge sails from a river into the sea. Does its draft increase or decrease, and by what fraction? Why is this still true if the barge is empty?
2. **Numerical.** The same barge (20 x 6 x 2.5 m, empty 60 t) carries 100 t of cargo in sea water. Find the draft.
3. **Graph and simulation interpretation.** In the draft graph, the sea-water line meets the red deck line at a larger cargo than the fresh-water line. Read both values and say what the difference means.
4. **Design and optimisation challenge.** Design: a raft is 3.0 m long, 0.25 m deep and has a mass of 10 kg per square metre of deck. It must carry 15 people of 65 kg in fresh water and keep a freeboard of at least 0.10 m. What is the narrowest width, in steps of 0.1 m? Why might you choose a wider one?

3 Session 3: Does it stay upright? Metacentric height and the righting arm

A. Real-world problem

A barge with plenty of freeboard can still roll over when its cargo is piled too high. Floating is about weight and buoyancy balancing; staying upright is about what happens when something nudges the barge over. How can we test it, and how high may we stack the cargo? (Educational 2D model only: not a naval architecture tool; real ships must be assessed by qualified designers using the applicable rules.)

B. Concept and theory

The weight of the barge acts downward at the centre of gravity G. The buoyant force acts upward at the centre of buoyancy B, which is the centre of the displaced water. Upright, both are on the centreline. When the barge heels by an angle phi the submerged shape changes, B moves toward the low side, and the two forces are a horizontal distance GZ apart: a righting arm of length GZ, giving a righting moment W * GZ. The metacentre M is the point where the buoyant force line crosses the centreline for small heel; if M is above G (GM $>$ 0) the moment rights the barge, if below (GM $<$ 0) it heels it further. Measuring heights up from the keel K, for a box KB = d / 2 and BM = I / V where I = L B^3 / 12 is the second moment of the water-plane area and V = L B d is the displaced volume, so BM = B^2 / (12 d). Then GM = KB + BM - KG. For a loaded barge KG is the mass-weighted mean of the heights of the empty barge and the cargo, so stacking cargo high raises KG and lowers GM. For larger angles we compute GZ exactly: rotate the rectangle, find the horizontal water line that leaves the same submerged area, and take the centroid of the wet polygon. For a box with the deck edge dry and the bilge wet the exact result is the wall-sided formula GZ = sin(phi) (GM + 0.5 BM tan^2(phi)).

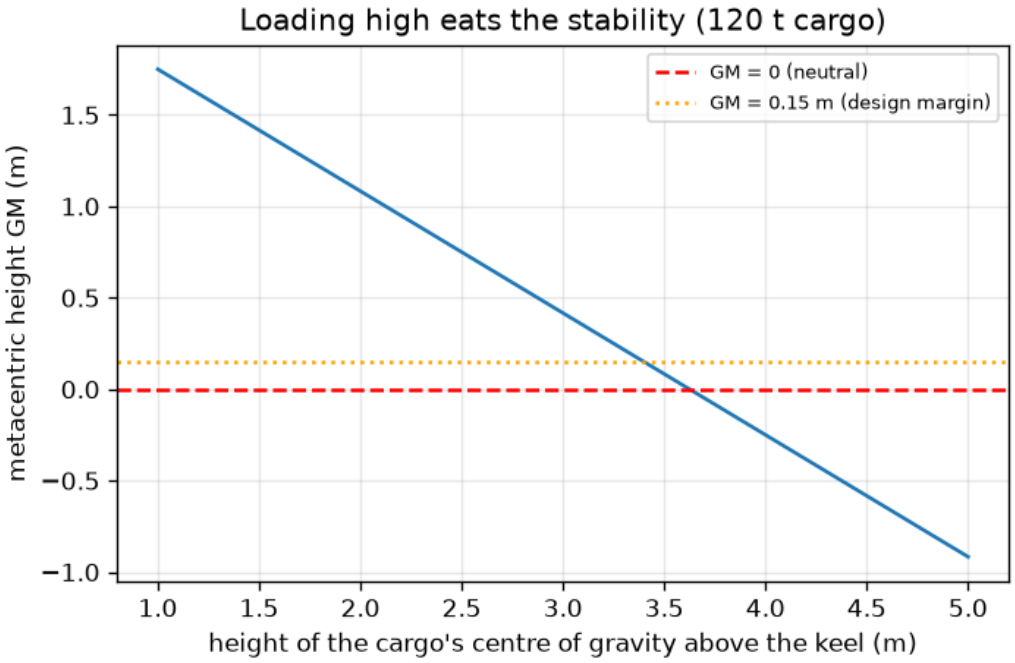
- K, B, G, M: keel, centre of buoyancy, centre of gravity, metacentre (heights above the keel are KB, KG, KM) (m)
- d: draft (m)
- I: second moment of the water-plane area, L B^3 / 12 (m^4)
- V: displaced volume (m^3)
- BM: metacentric radius I / V (m)
- GM: metacentric height KB + BM - KG (m)
- phi: heel angle (degrees or radians)
- GZ: righting arm (m)

$$KB = \frac{d}{2}, \qquad BM = \frac{I}{V} = \frac{LB^3/12}{LBd} = \frac{B^2}{12\,d}$$

$$GM = KB + BM - KG, \qquad KG = \frac{m_0\,KG_0 + m_c\,h}{m_0 + m_c}$$

$$GZ = y_G - y_B, \qquad GZ \approx GM \sin\varphi \ \text{(small angle)}$$

$$GZ = \sin\varphi\left(GM + \tfrac{1}{2}BM\tan^2\varphi\right) \ \text{(box, deck edge dry, bilge wet)}$$



Metacentric height against the height of the cargo's centre of gravity.

E. Worked example

The barge (20 x 6 x 2.5 m, empty 60 t with its centre of gravity 1.0 m above the keel) carries 120 t of cargo whose centre of gravity is 2.5 m above the keel, in fresh water. Find GM, and the righting arm at 10 degrees exactly and by the small-angle formula.

1. Total mass 180 000 kg, so draft d = 180 000 / (1000 * 20 * 6) = 1.500 m.
2. KB = d / 2 = 0.750 m. BM = B^2 / (12 d) = 36 / 18 = 2.000 m. KM = 2.750 m.
3. KG = (60 * 1.0 + 120 * 2.5) / 180 = 360 / 180 = 2.000 m.
4. GM = 2.750 - 2.000 = 0.750 m, positive, so the barge rights itself.
5. Small angle at 10 degrees: GM sin(10 deg) = 0.750 * 0.1736 = 0.1302 m.
6. Wall-sided formula: sin(10 deg) (0.750 + 0.5 * 2.000 * tan^2(10 deg)) = 0.1736 * (0.750 + 0.0311) = 0.1356 m, which equals the exact numerical value 0.1356 m; the small-angle value is 4 percent low.

Answer. GM = 0.750 m (stable). At 10 degrees the exact righting arm is 0.1356 m and the small-angle estimate is 0.1302 m.

F. Questions

1. **Conceptual.** A barge has 1.2 m of freeboard but capsizes when loaded with tall stacked cargo. Explain why freeboard does not guarantee stability, and what quantity does.
2. **Numerical.** The same barge carries 120 t with its centre of gravity at 3.0 m. Find KG and GM, and say whether it is stable.
3. **Graph and simulation interpretation.** From the GZ figure, at 20 degrees the exact curve is above the small-angle line. Read approximately both values for the loaded barge, state which is larger and give a likely reason for the difference.
4. **Design and optimisation challenge.** Design: with 160 t of cargo (instead of 120 t), find the highest cargo centre of gravity, in steps of 0.1 m, that keeps GM at or above 0.15 m. Compare it with the 3.40 m for 120 t and explain.

4 Session 4: The free-surface effect: why a liquid in a tank lowers GM

A. Real-world problem

In Session 3 the 180 t barge (60 t empty, 120 t of cargo with its centre 2.5 m above the keel) has a healthy GM of 0.75 m. Now the 120 t is fresh water in a tank 10 m long and 5 m wide. When the barge heels, a solid cargo stays put, but a liquid with a free surface runs to the low side and pulls the barge further over. How much of the 0.75 m is lost, why does a slightly wider tank turn a stable barge into an unstable one, and what does a divider do? (Educational 2D model only: not for loading a real vessel.)

B. Concept and theory

The tank. A rectangular tank of length l (along the barge) and width w (across), on the centre line, holds liquid of density rho_l and mass m_l, so the liquid is h = m_l / (rho_l l w) deep.

The loss. When the barge heels by a small angle phi the liquid surface stays level, so a wedge of liquid moves to the low side: the liquid's centre of gravity shifts sideways by i phi / v, where i = l w^3 / 12 is the second moment of the free surface and v the volume of the liquid. This shifts the barge's total weight sideways exactly as if its centre of gravity G had been raised by

FSC = rho_l i / m,

where m is the total mass of the barge (the same as rho times the displaced volume). The effective metacentric height is GM_eff = GM_solid - FSC = KM - (KG + FSC). Two things to notice. FSC does not depend on how full the tank is (as long as the surface does not touch the roof or the floor): the liquid's weight and the size of the wedge cancel. And it grows with the cube of the width.

Dividers. A longitudinal divider (along the barge) splits the width into n compartments of width w / n; each has i = l (w/n)^3 / 12, so together they have i / n^2. Two compartments cut the loss to a quarter and three to a ninth. A transverse bulkhead (across the barge) shortens the compartments but leaves each as wide as the tank, so the total l_j w^3 / 12 adds up to the same i: it does not help. A completely full tank has no free surface and FSC = 0.

Large heel. The formula is the small-angle limit. The exact GZ curve (liquid centroid from the heeled tank geometry) falls below the small-angle line GM_eff sin(phi) as the heel grows.

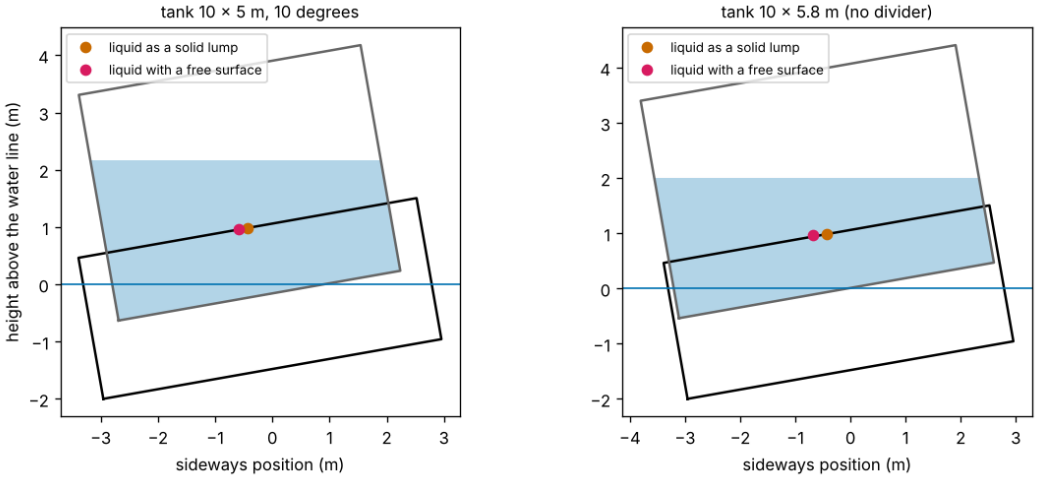
- l, w, h: tank length, width and liquid depth (m)
- rho_l: density of the liquid (1000 for fresh water) (kg/m^3)

- i: second moment of the free surface, l w^3 / 12 (m^4)
- m: total mass of the barge and cargo (kg)
- FSC: free-surface correction, rho_l i / m (m)
- GM_solid, GM_eff: metacentric height with the liquid as a solid, and with the free surface (m)
- n: number of equal longitudinal compartments (no unit)
- GZ: righting arm (m)

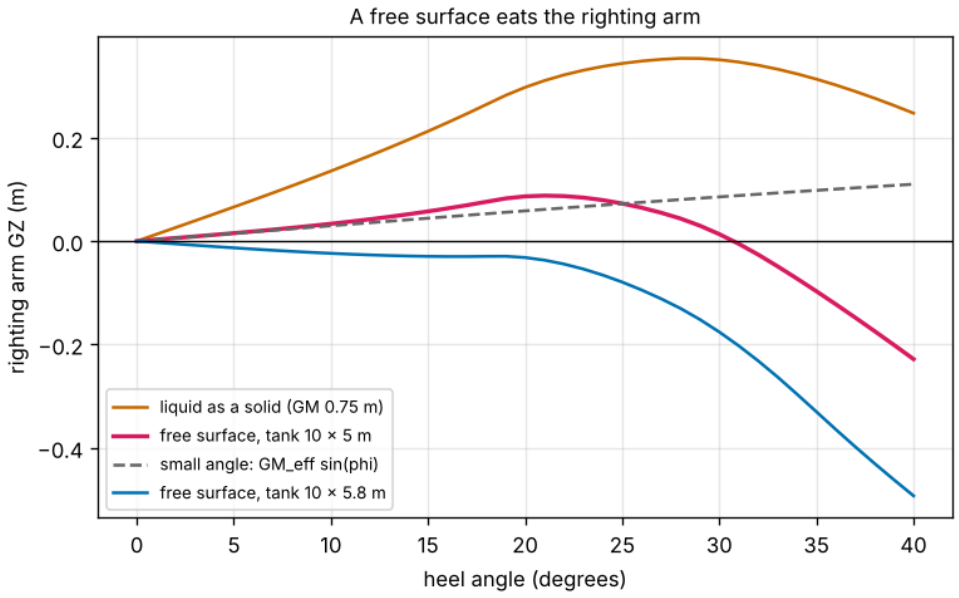
$$i = \frac{l\,w^3}{12}, \qquad \mathrm{FSC} = \frac{\rho_l\, i}{m}, \qquad GM_{\mathrm{eff}} = GM_{\mathrm{solid}} - \mathrm{FSC}$$

$$i_n = n \cdot \frac{l\,(w/n)^3}{12} = \frac{i}{n^2} \;\; (n \text{ longitudinal compartments}), \qquad \sum_j \frac{l_j\, w^3}{12} = i \;\; (\text{transverse bulkheads})$$

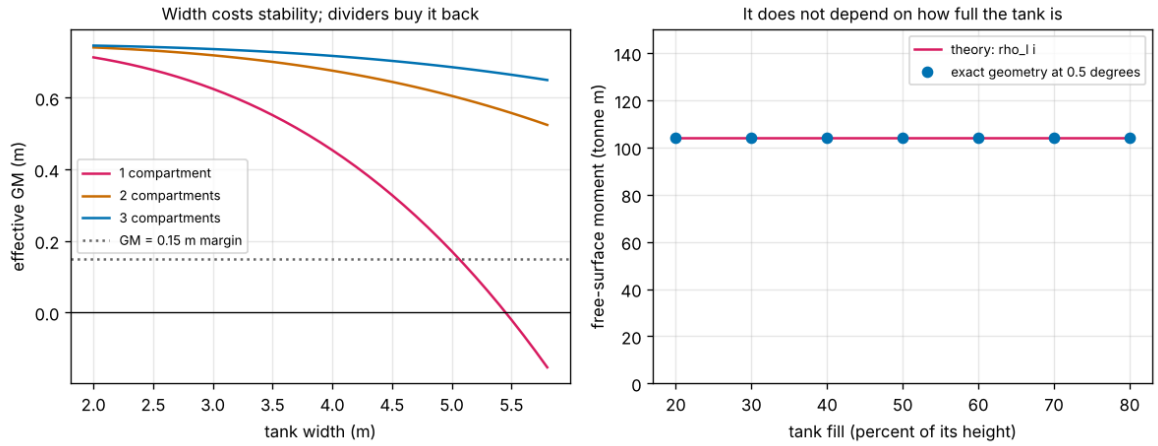
$$GZ = y_G - y_B, \qquad y_G = \frac{m_0\, y_0 + m_l\, y_{\mathrm{liquid}}}{m_0 + m_l} \;\; (y_{\mathrm{liquid}} \text{ from the heeled tank})$$



The heeled barge with its tank: the liquid with a free surface (magenta) against the same liquid frozen as a solid lump (orange).



Righting arm against heel angle: solid cargo, the 10 x 5 m tank, the small-angle line and the 5.8 m wide tank.



Left: effective GM against tank width for 1, 2 and 3 compartments. Right: the free-surface moment against the fill of the tank.

E. Worked example

Find the effective GM of the 180 t barge when its 120 t cargo is fresh water in a tank 10 m long and 5 m wide (centre of the liquid 2.5 m above the keel), and then with one longitudinal divider. Check the first value with the exact heeled geometry.

1. The solid-cargo GM is 0.750 m (Session 3, worked example: KB 0.750, BM 2.000, KG 2.000).
2. The free-surface second moment is i = l w^3 / 12 = 10 x 125 / 12 = 104.17 m^4.
3. The correction is FSC = rho_l i / m = 1000 x 104.17 / 180 000 = 0.5787 m.
4. The effective GM is 0.750 - 0.5787 = 0.1713 m: still positive, but nearly neutral.
5. With one divider each compartment is 2.5 m wide, so i = 2 x 10 x 2.5^3 / 12 = 26.04 m^4 (a quarter), FSC = 0.1447 m and GM_eff = 0.6053 m.
6. The exact heeled geometry at 0.5 degrees gives GM_eff = 0.1714 m and 0.6054 m, which agree with the formula to 0.0001 m.

Answer. GM_eff = 0.1713 m for the undivided 10 x 5 m tank (the barge loses 77 percent of its GM; the roadmap asked for 0.17 m or less, which is this value to two decimals) and 0.6053 m with one divider.

F. Questions

1. **Conceptual.** Why does the free-surface loss grow with the cube of the tank width, and why does a transverse bulkhead not help while a longitudinal divider does?
2. **Numerical.** A tank 8 m long and 4 m wide holds fresh water on a barge of total mass 150 t whose solid-cargo GM is 0.60 m. Find i, the free-surface correction and the effective GM, and then the effective GM with one longitudinal divider.
3. **Graph and simulation interpretation.** In the righting-arm figure the small-angle line GM_eff sin(phi) gives 0.0856 m at 30 degrees for the 10 x 5 m tank, but the exact curve gives 0.0134 m, while the solid-cargo curve gives 0.3510 m. What do you conclude about the formula, and about the range of stability?
4. **Design and optimisation challenge.** The owner wants an effective GM of at least 0.15 m with the 120 t of liquid in a tank 10 m long. What is the widest undivided tank, and what is the cheapest change that lets the full 5.8 m width be used?

5 Real-world mini-project

How high can the deck cargo go? A barge 20 m long and 6 m wide (hull depth 2.5 m, empty mass 60 t, empty centre of gravity 1.0 m above the keel) will carry 120 t of cargo in fresh water. Two rules must hold: the metacentric height must be at least 0.15 m, and if the cargo slips 0.2 m sideways the barge must heel by no more than 10 degrees. Use `gm_loaded` and `heel_after_shift` from Session 3 and find the highest cargo centre of gravity (in steps of 0.05 m) that meets each rule and both together.

How wide may the tank be, and does it need a divider? The barge carries 120 t of fresh water (centre 2.5 m above the keel) in a tank 10 m long. The tank width can be 4.0, 4.5, 5.0, 5.5 or 5.8 m, with 1, 2 or 3 longitudinal compartments of equal width. A choice passes if the effective GM is at least 0.15 m. Use `make_tank`, `gm_effective` and `max_tank_width` from Session 4. For the undivided tank 5.8 m wide, also use `righting_arm_tank` to say what the barge does after a nudge. **Deliverable:** A script that prints the 15 effective GM values with pass or fail, reports the widest undivided tank that passes, reports the fewest compartments that let the full 5.8 m width pass, and saves a plot of effective GM against width for 1, 2 and 3 compartments with the margin marked. Add two sentences on why a barge that is stable with a solid cargo can capsize with a liquid. **Success looks like:** The undivided tank passes for 4.0, 4.5 and 5.0 m (GM 0.454, 0.328 and 0.171 m) and fails for 5.5 m (-0.020 m) and 5.8 m (-0.153 m); the widest undivided width is 5.06 m. With 2 or 3 compartments all five widths pass (0.524 to 0.717 m), so one divider is enough for the full 5.8 m. For the undivided 5.8 m tank the righting arm is negative at every heel angle from 1 to 60 degrees (-0.024 m at 10 degrees), so the model has no angle of loll and the barge keeps heeling.

6 The path ahead

1. **Session 5: A round-bilge hull.** A generalised wet-section function for any polygon hull, with the GZ curve plotted.
2. **Session 6: Roll in time.** A time-stepping model of rolling that uses the GZ curve as the restoring moment.
3. **Session 7: Settle at the angle of loll.** A dynamic simulation of the failure case with a small disturbance.
4. **Session 8: A stability report for a loading plan.** A script that reads cargo items, finds KG, GM, the GZ curve and the area under it, and prints pass or fail.

7 Answers and hints

Session 1

1. **Conceptual.** Hint: Think of the average density of the whole ship including the air inside. Solution: The ship is a hollow shell, so its total mass divided by its total outside volume (steel plus the air inside) is far below 1000 kg/m^3. In Python, a 150 kg steel boat of outside volume 0.8 m^3 has an average density of 187.5 kg/m^3 and floats, while the same 150 kg of steel as a lump has volume 0.019 m^3 and density 7850 kg/m^3 and sinks.
2. **Numerical.** Hint: Average density = mass / volume; compare with 1000. Solution: Density = 2.0 / 0.0030 = 666.7 kg/m^3, below 1000, so it floats. Fraction under water = 666.7 / 1000 = 0.667. It pushes aside 2.0 kg of water, the same as its own mass (check: 1000 * 0.0030 * 0.667 = 2.0 kg).
3. **Graph and simulation interpretation.** Hint: Find 1010 on the horizontal axis for each water and see whether the fraction reaches 1. Solution: In fresh water the fraction would be

1010 / 1000 = 1.01, which is more than 1, so it sinks. In sea water (1025) the fraction is 1010 / 1025 = 0.985, so it floats with 98.5 percent under water, just barely. Sea water gives a line that reaches 1 further to the right.

4. **Design and optimisation challenge.** Hint: Largest total mass = rho * A * draft_max. Solution: Bottom area = 1.6 m^2. Largest total mass = 1000 * 1.6 * 0.25 = 400 kg, so the cargo is at most 400 - 150 = 250 kg. Assumptions: calm water, the boat stays upright, the cargo is spread evenly and the sides are tall enough. A real boat needs a bigger safety margin (waves, uneven loading), and Session 3 shows that stability can fail even when the boat floats.

Session 2

1. **Conceptual.** Hint: Displaced water must weigh the same; compare the densities. Solution: The barge must displace the same mass of water, and sea water is denser, so the volume displaced is smaller and the draft decreases. The ratio is 1000 / 1025 = 0.976, so the draft drops by about 2.4 percent. It holds for any load, including the empty barge, because the rule is draft = m / (rho L B).
2. **Numerical.** Hint: draft = m / (rho L B) with rho = 1025. Solution: Draft = 160 000 / (1025 * 20 * 6) = 160 000 / 123 000 = 1.301 m, a little less than 1.333 m in fresh water. Check units: kg / (kg/m^3 * m * m) = m.
3. **Graph and simulation interpretation.** Hint: The deck line is at 2.5 m draft; solve m = rho L B H. Solution: Fresh water: 240 t; sea water: 1025 * 20 * 6 * 2.5 / 1000 - 60 = 247.5 t. The difference is 7.5 t, which is the extra 2.5 percent of displaced mass that denser water can support at the same volume.
4. **Design and optimisation challenge.** Hint: Capacity = rho * L * B * (H - f_min) - raft mass, where raft mass = 10 * L * B. Solution: Capacity = 1000 * 3 * b * 0.15 - 30 * b = 420 b kg. For 15 people we need 975 kg, so b >= 975 / 420 = 2.32 m, and the narrowest width in steps of 0.1 m is 2.4 m (capacity 1008 kg, 15 people; at 2.3 m the capacity of 966 kg carries only 14). A wider raft gives a reserve for waves, uneven standing and heavier people, and it is also more stable (Session 3). The trade-off is more material, cost and weight to handle.

Session 3

1. **Conceptual.** Hint: Compare the height of G with that of M. Solution: Freeboard says how far the deck is above the water; it comes from weight and buoyancy balancing (Session 2). Stability depends on where G is relative to M. Tall cargo raises G (KG) and lowers GM = KM - KG; if G rises above M, GM is negative and a small heel is pushed further by the water instead of being corrected, regardless of how much freeboard there is.
2. **Numerical.** Hint: KG = (60 * 1.0 + 120 * 3.0) / 180; KM is still 2.750 m. Solution: KG = (60 + 360) / 180 = 2.333 m. GM = 2.750 - 2.333 = 0.417 m. It is stable but with less than the 0.750 m of the worked example, and it is above the 0.15 m margin. Units: all heights in metres.
3. **Graph and simulation interpretation.** Hint: The deck edge goes under at 18.4 degrees; compare 0.298 and 0.257. Solution: At 20 degrees the exact GZ is 0.298 m and GM sin(phi) is 0.257 m: the exact curve is about 14 percent higher. The small-angle formula assumes the centre of buoyancy moves as if the waterplane stayed a rectangle; the BM tan^2(phi) term of the wall-sided formula adds the extra shift of B as the submerged shape becomes a wedge. By 20 degrees the deck edge is just under water, so even the wall-sided formula (0.302 m) starts to overestimate.
4. **Design and optimisation challenge.** Hint: Total 220 t gives draft 220 000 / 120 000 = 1.833 m; find KM and then KG_max. Solution: Draft = 1.833 m, KB = 0.917 m, BM = 36 / (12 * 1.833) = 1.636 m, KM = 2.553 m. KG_max = 2.553 - 0.15 = 2.403 m, so 60 + 160 h <= 2.403 * 220 = 528.7 and h <= 2.93 m; in steps of 0.1 m the answer is 2.9 m (GM =

0.171 m; at 3.0 m GM is 0.099 m, which fails). More cargo gives a lower height because it raises KG more and, since the draft rises, BM falls. Trade-off: carrying more means stowing it lower, e.g. in the hold, or accepting a smaller margin, which is not wise.

Session 4

1. **Conceptual.** Hint: Think about the second moment of the free surface, i = l w^3 / 12, and how each kind of wall changes l and w. Solution: The loss is rho_l i / m and i = l w^3 / 12: the liquid wedge that moves when the barge heels is as wide as the surface and moves a distance proportional to the width, so its moment grows with w^3 (a tank twice as wide has 8 times the loss). A longitudinal divider replaces one surface of width w by n surfaces of width w / n, each with i = l (w/n)^3 / 12, so the total is i / n^2: 1/4 for two compartments. A transverse bulkhead splits the length l into pieces l_j, but each piece is still w wide and the sum of l_j w^3 / 12 is the same i: the loss is unchanged.
2. **Numerical.** Hint: i = l w^3 / 12, FSC = 1000 i / m with m in kg, and one divider gives i / 4. Solution: i = 8 x 4^3 / 12 = 42.67 m^4. FSC = 1000 x 42.67 / 150 000 = 0.2844 m, so GM_eff = 0.60 - 0.2844 = 0.3156 m. With one divider i = 10.67 m^4, FSC = 0.0711 m and GM_eff = 0.5289 m. Units: m^4 for i, metres for the correction.
3. **Graph and simulation interpretation.** Hint: The formula is a small-angle result. Compare the three numbers and look at where each curve reaches zero. Solution: The formula FSC = rho_l i / m is the limit of zero heel: at 5 degrees the line gives 0.0149 m and the exact curve 0.0154 m, but at 30 degrees the line overestimates the righting arm by 0.072 m, because the liquid wedge keeps growing and moves the weight further to the low side. The exact curve, not the line, must be used at large angles. The solid-cargo curve is positive up to 53.6 degrees, while the free-surface curve reaches zero at 30.7 degrees: the free surface also shortens the range of stability by 23 degrees, which the single number GM_eff does not show.
4. **Design and optimisation challenge.** Hint: Use i_max = (GM_solid - 0.15) m / rho_l and w = (12 i_max / l)^(1/3); then try n = 2 compartments. Solution: i_max = (0.75 - 0.15) x 180 000 / 1000 = 108 m^4, so w^3 = 12 x 108 / 10 = 129.6 and w = 5.06 m: a 5.0 m tank passes (GM 0.171 m) but 5.5 m does not (-0.020 m). With n compartments the widest width is n (12 i_max / (n l))^(1/3): 8.03 m for two and 10.53 m for three, both more than the hull can take, so one longitudinal divider is enough for the full 5.8 m width (GM_eff 0.524 m). A divider costs far less than narrowing the tank, and it also lengthens the range of stability (GZ vanishes at 42.2 degrees against 30.7 for the 10 x 5 m tank).

8 How to join

Read `CONTRIBUTING.md` in the repository, then pick an issue labelled `good first issue`. Repository: https://github.com/KrunalPanchal1995/openbuoyancy